EEE 201

Muhammad Ahad Rahman Miah
Assistant Professor, EEE
AHAD@uap-bd.edu

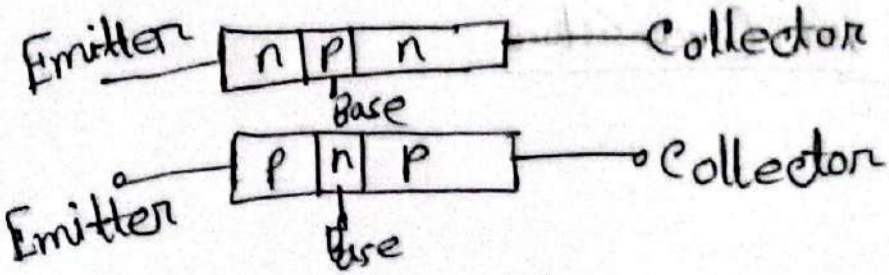
- BJT (Bipolar Junction Transistor)
- FET (Field Effect Transistor)

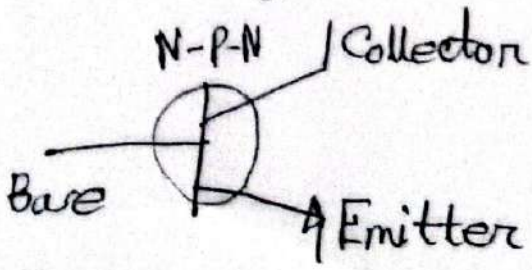
Transfer Varistor থেকে Transistor নামের এসেছে।

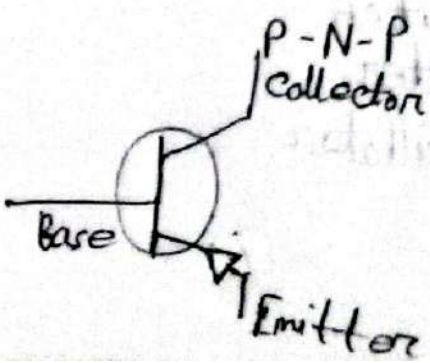
- A transistor is a device that acts depending on the voltage level of the input signal, either as a conductor or as a resistor of the flow of electricity.

Conductor, Semiconductor, Isolator

BJT
- PNP Transistor
- NPN Transistor



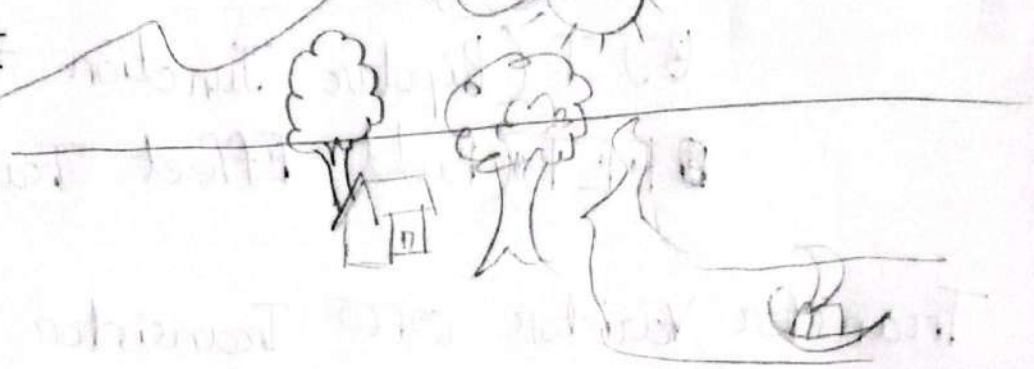




3/8/2026

Emitter: It is a region situated in one side of a a transistor, which supplies charge carriers to the other two regions

Base: I



Bipolar

Both polarity electrons and holes Participate in the conduction process

Junction

It has two P-N junction (CB & EB)

Transistor

It has transfer resistor. Its resistance changes with respect to region of operation.

Application of BGT (Transistor

1. Amplifier
2. Switch
3. Fitter
4. Oscillator

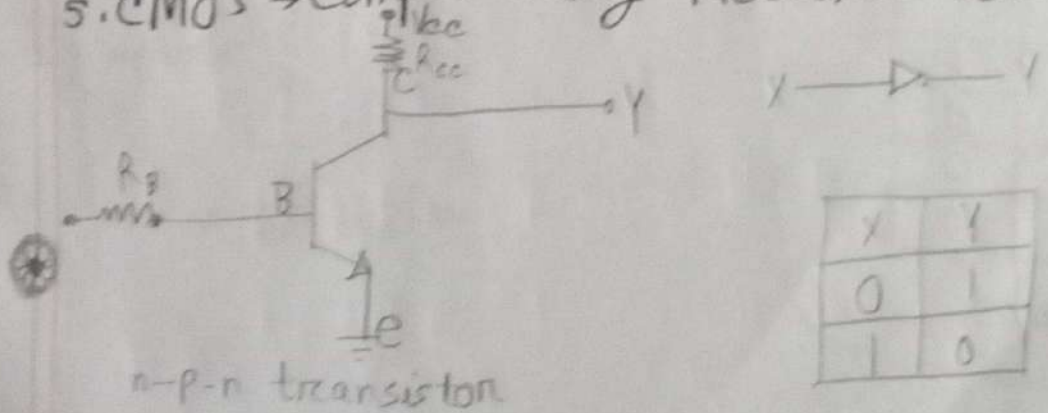
Advantages of BGT (Transistor)

1. High Voltage gain
2. Lower voltage supply
3. Lower power dissipation
4. Small size

Disadvantages of BGT (Transistor)

1. Temparature dependence
2. Lower switching speed

1. RTL → Resistor Transistor Logic
2. DTL → Diode Transistor Logic
3. TTL → Transistor Transistor Logic
4. NMOS → N-Channel Metaloxide Semi-conductor
5. CMOS → Complementary Metaloxide Semi Conductor (NMOS + PMOS)

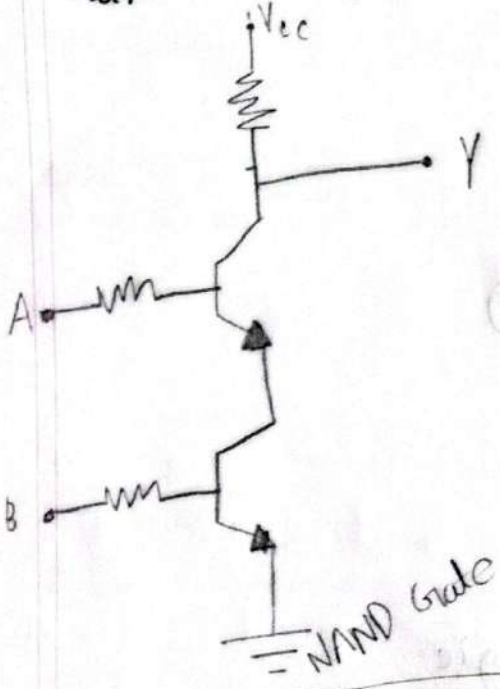


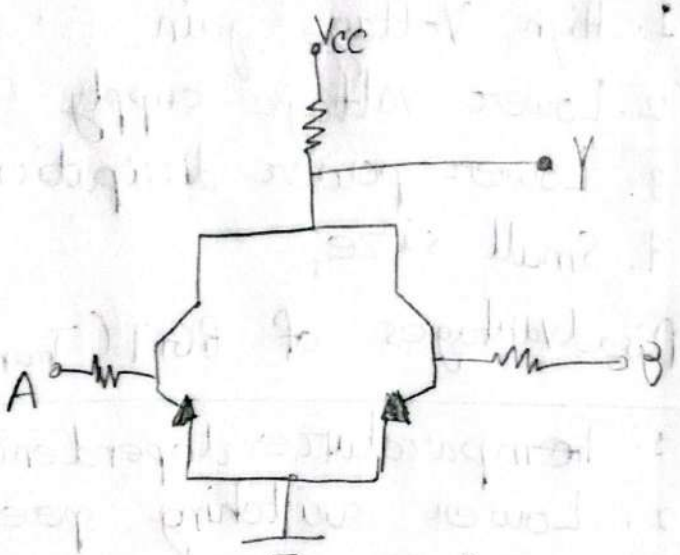
X	Y
0	1
1	0

NOT-Gate

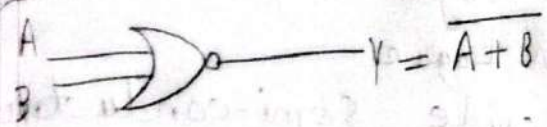
Product → Series

Sum → Parallal



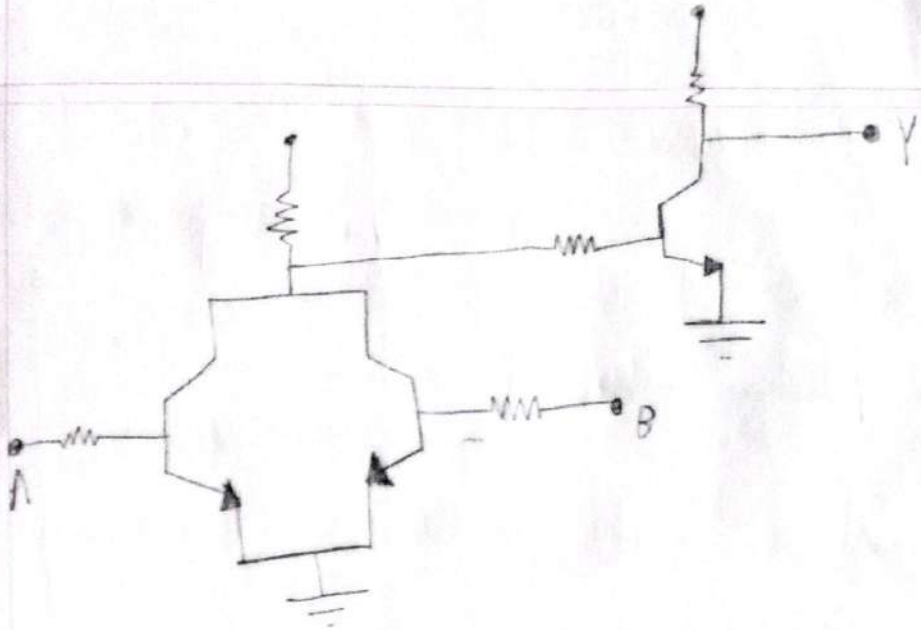


NOR Gate RTL

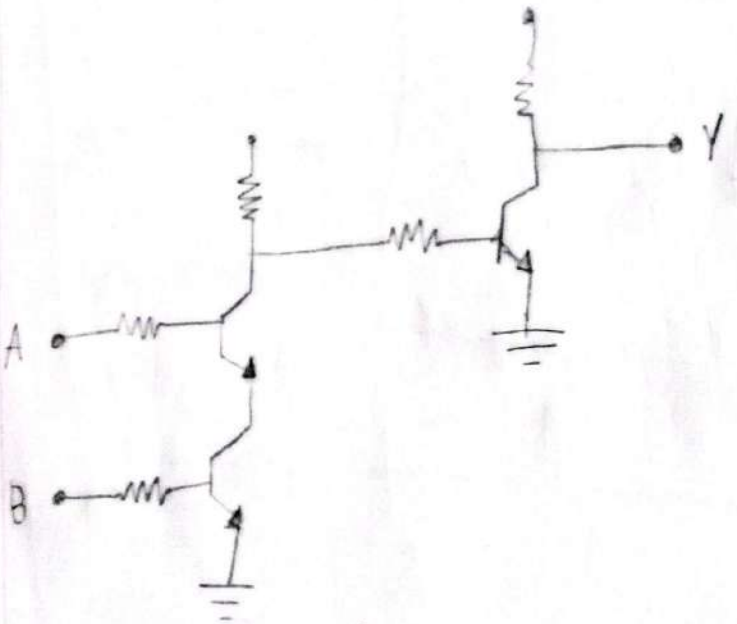


A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR Gate



OR Gate



AND Gate

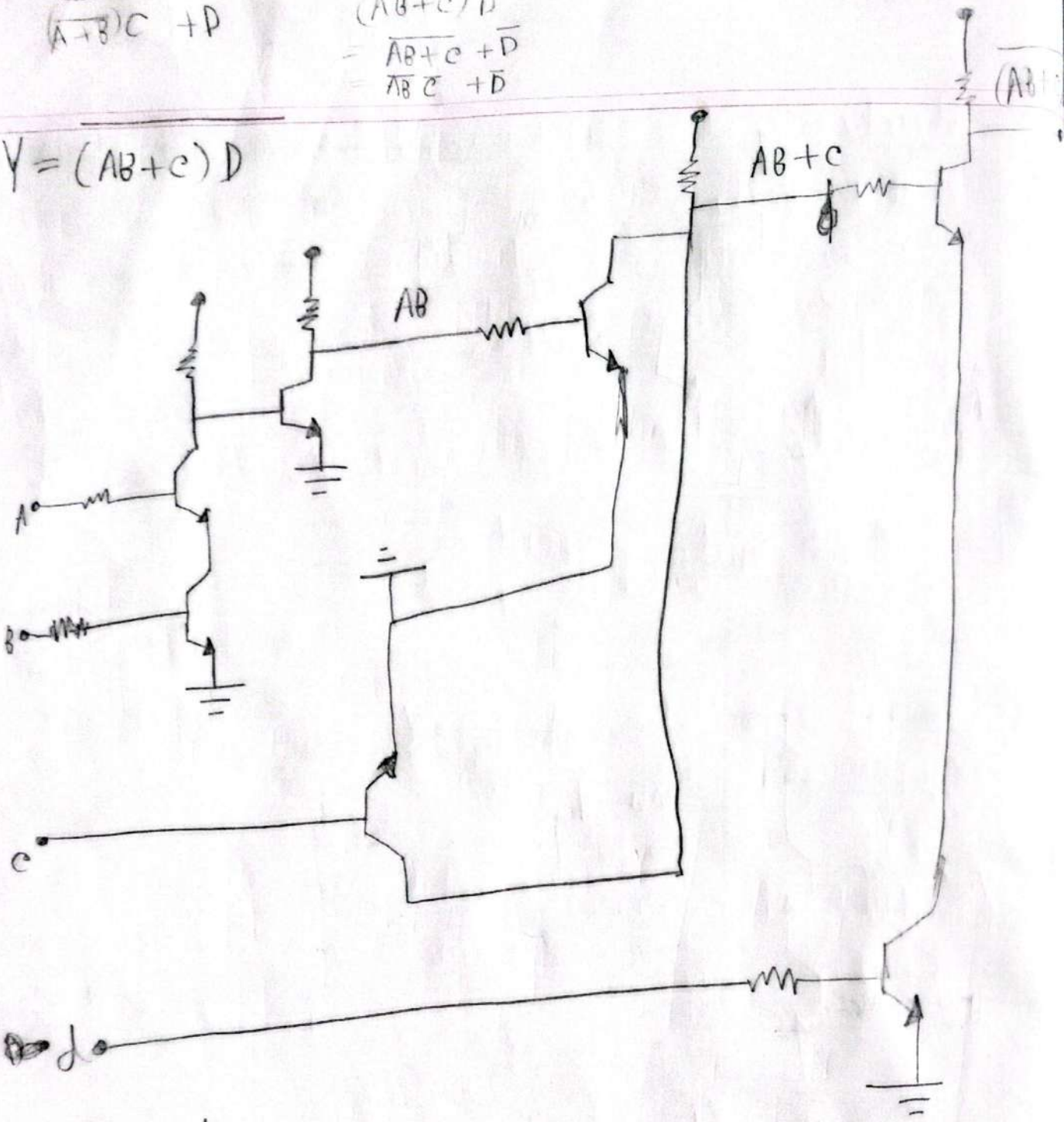
$\overline{(\overline{A+B})C} + D$

$(AB+C)D$

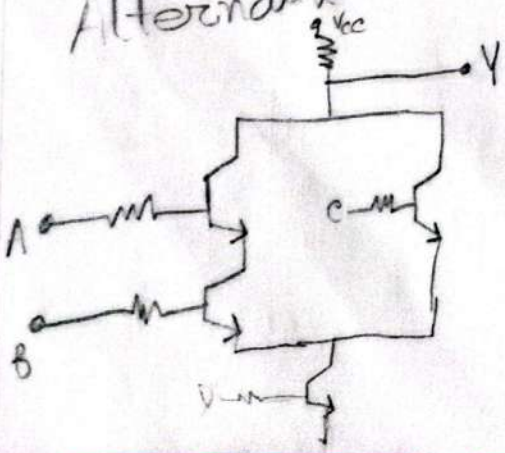
$= \overline{\overline{AB+C}} + \overline{D}$

$= \overline{\overline{AB}\,\overline{C}} + \overline{D}$

$Y = (AB+C)D$



Alternative



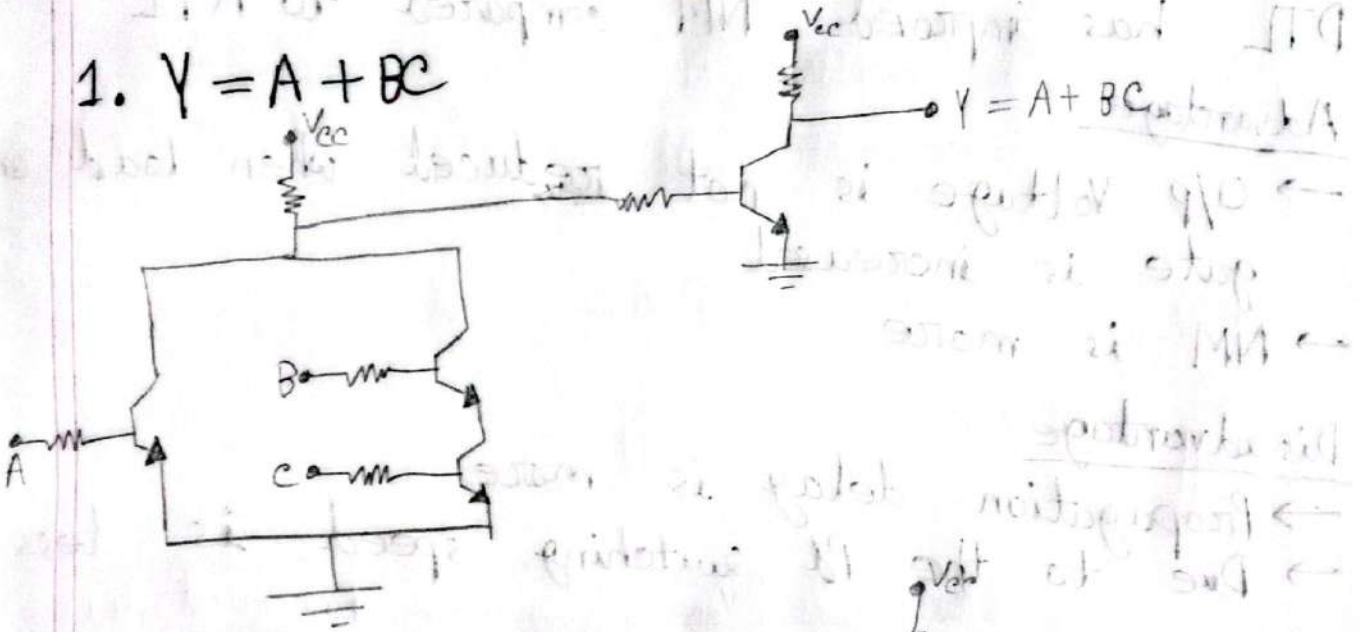
H.W

$Y = A + BC$ RTL Draw

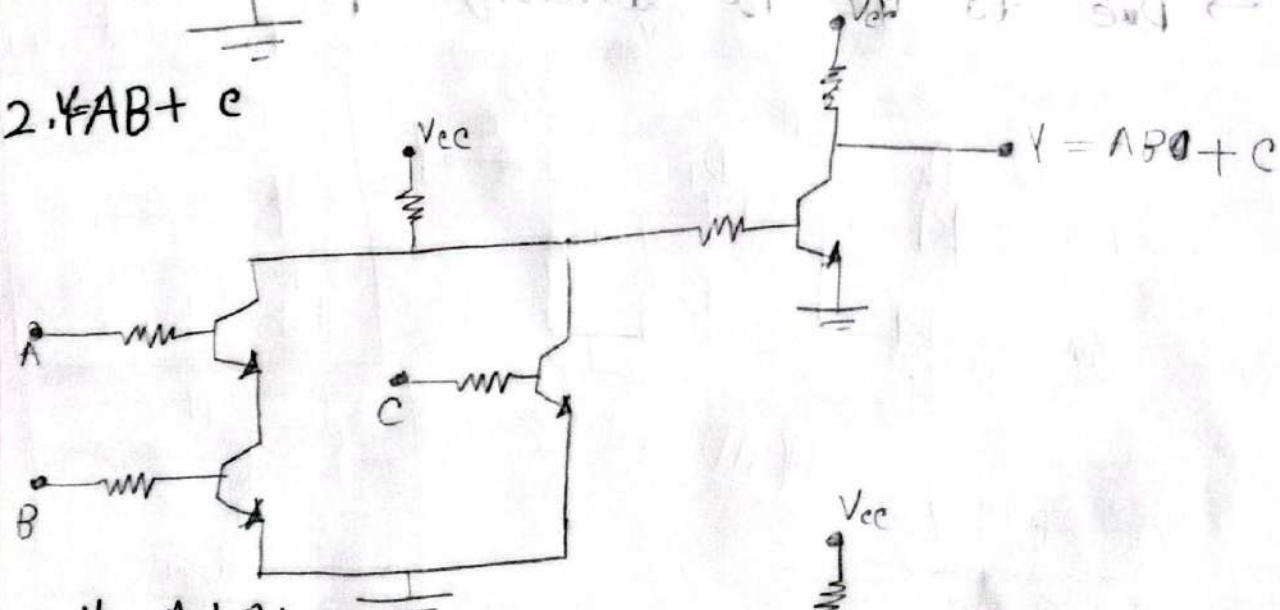
$Y = AB + C$

$Y = A + B + C$

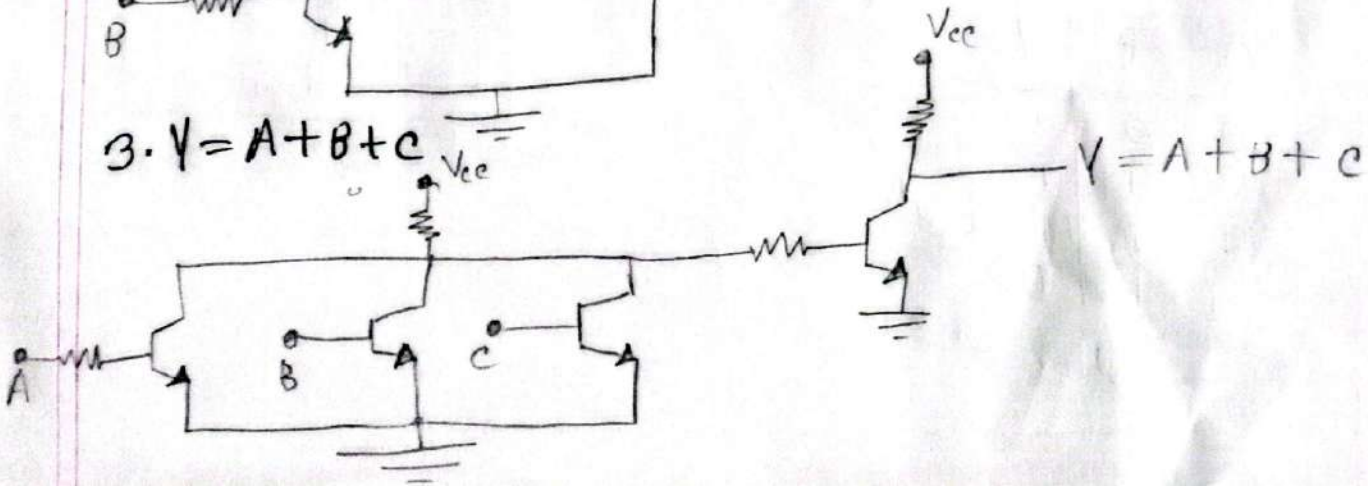
1. $Y = A + BC$



2. $Y = AB + C$



3. $Y = A + B + C$



9th August 2026

O/p - output

DTL

RTL has low noise margin, slow speed & higher Power disipation.

DTL has improved NM compared to RTL

Advantage

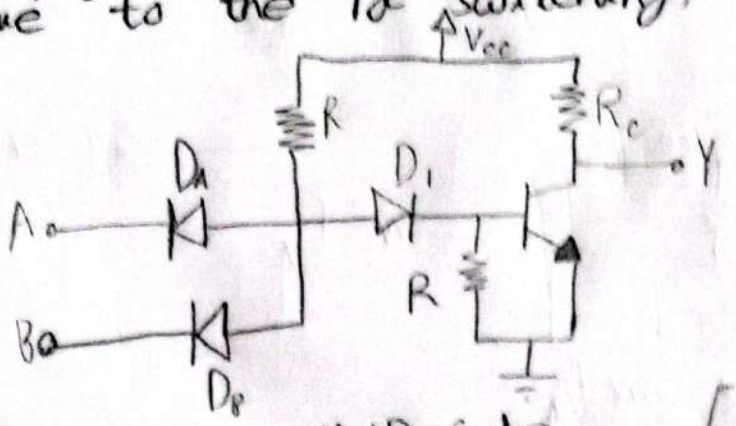
→ O/p Voltage is not reduced when load on gate is increased

→ NM is more

Disadvantage

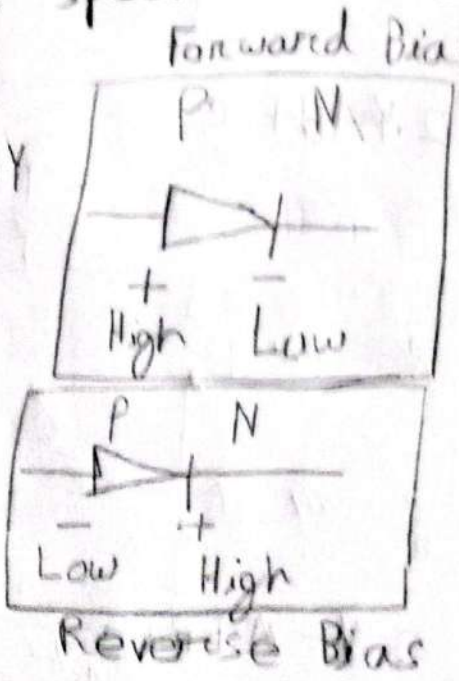
→ Propagation delay is more

→ Due to the Pd switching speed is less

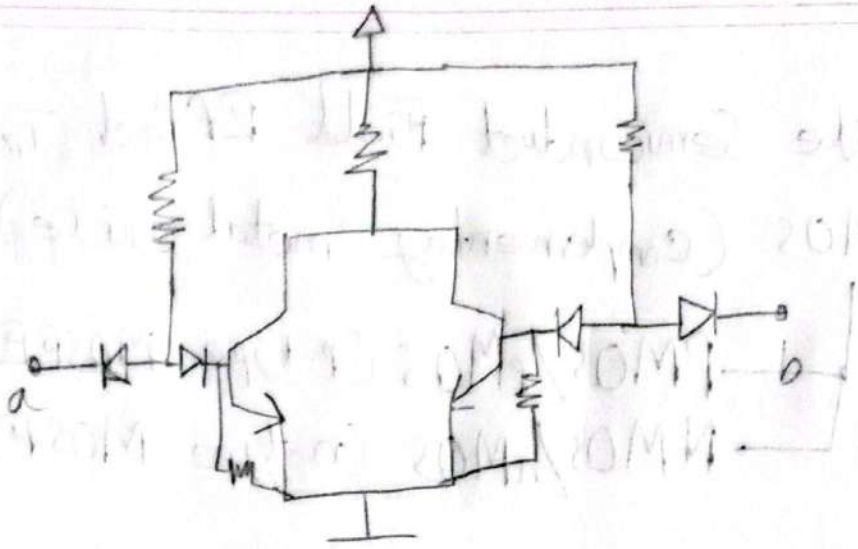


NAND Gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



next class: CMOS
PT CT



NOR Gate

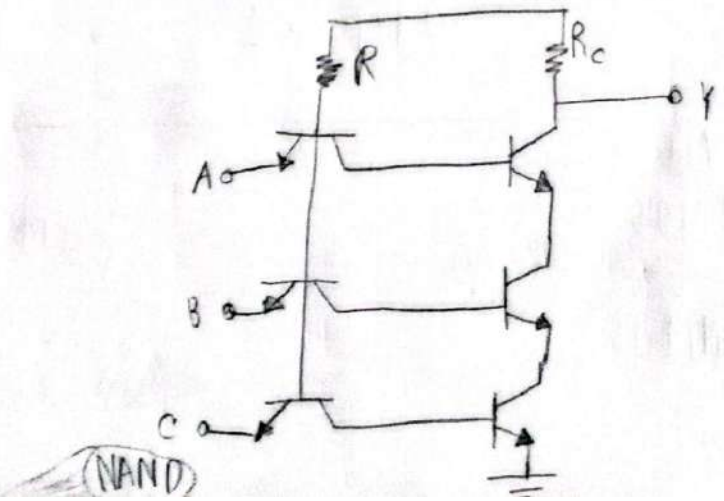
TTL

Advantage

- Circuit gives low o/p impedance
- Gate can drive capacitive load due to low o/p impedence
- power dissipation is less

Disadvantages

Large current spike is produced while switching



NAND Gate			
A	B	C	$\overline{ABC}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

10 August 2026

MOSFET

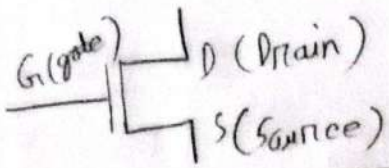
Metal Oxide Semiconduct Field Effect Transistor

└ cMOS (complementry metal oxide)

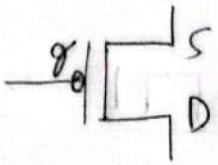
  ├ PMOS/pMOS (p-type MOSFET)

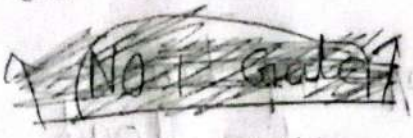
  └ NMOS/nMOS (n-type MOSFET)

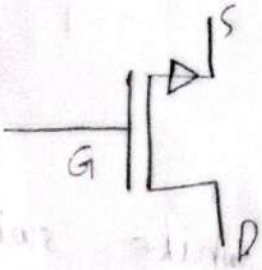
NMOS



PMOS

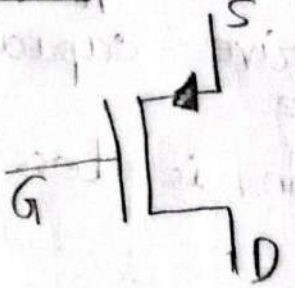






gate → 1

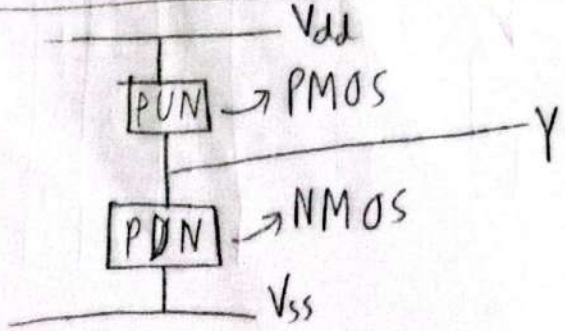
off → 0



hole না থাকলে current প্রবাহিত হবে

ON → 0

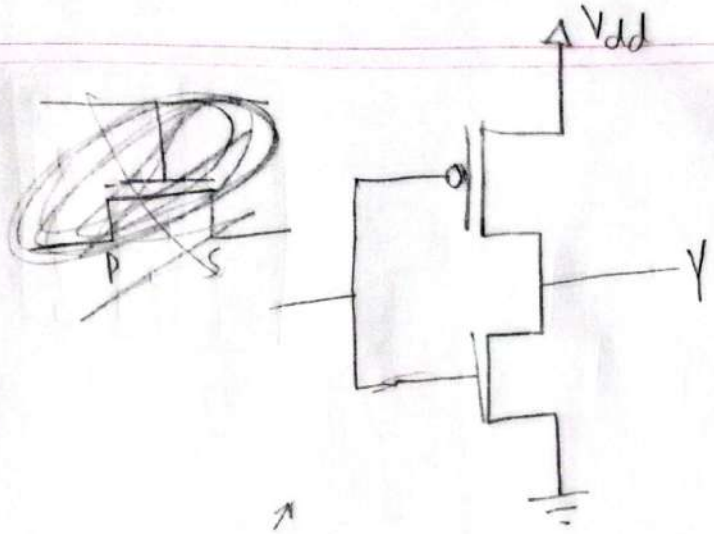
OFF → 1



PUN
Pull up Network

PDN
↓
Down

} cMOS এর structure



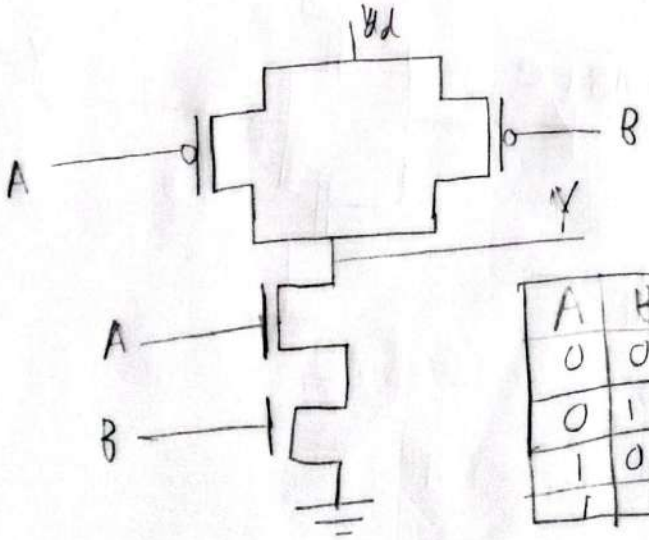
CMOS NOT Gate

2 Input Nand

Rule

Dot(•) হলে PMOS → Parallal
NMOS → Series

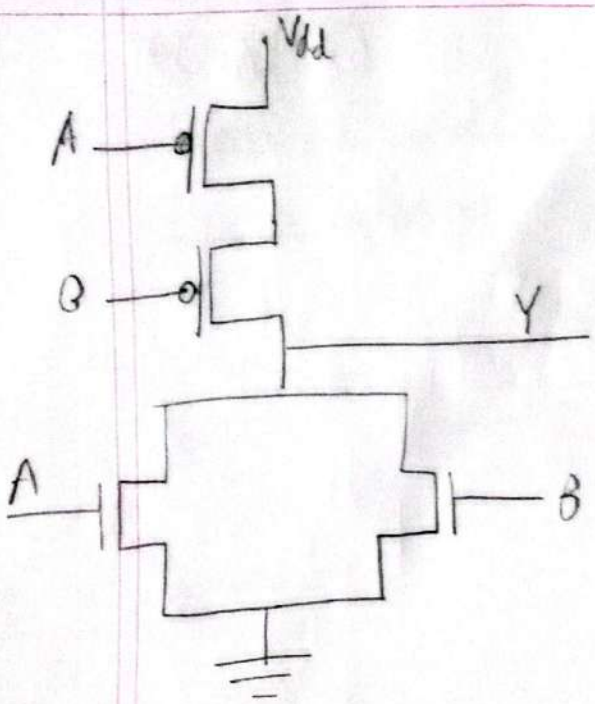
Plus(+) হলে PMOS → Series
NMOS → Parallal



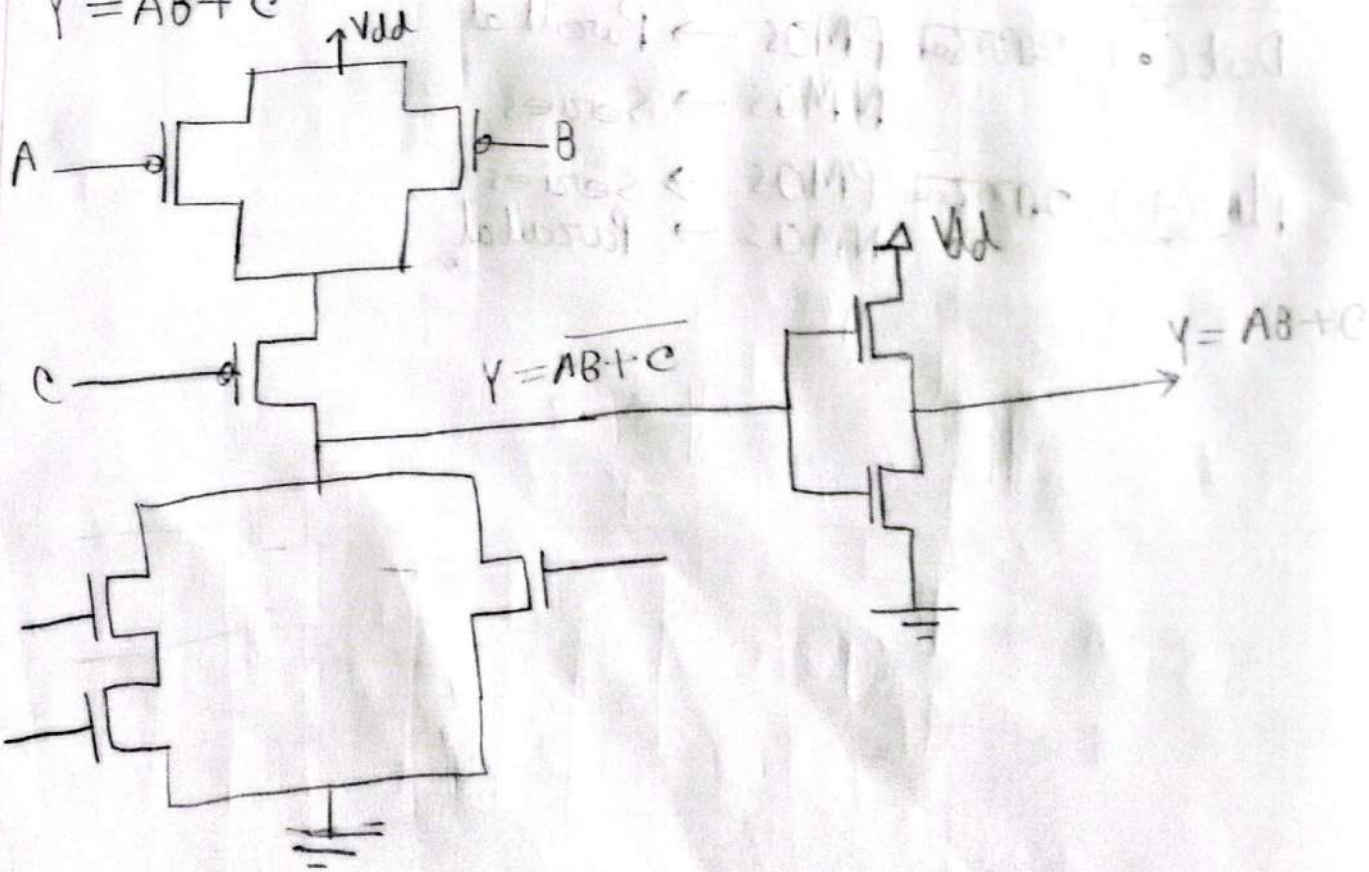
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

Nand Gate

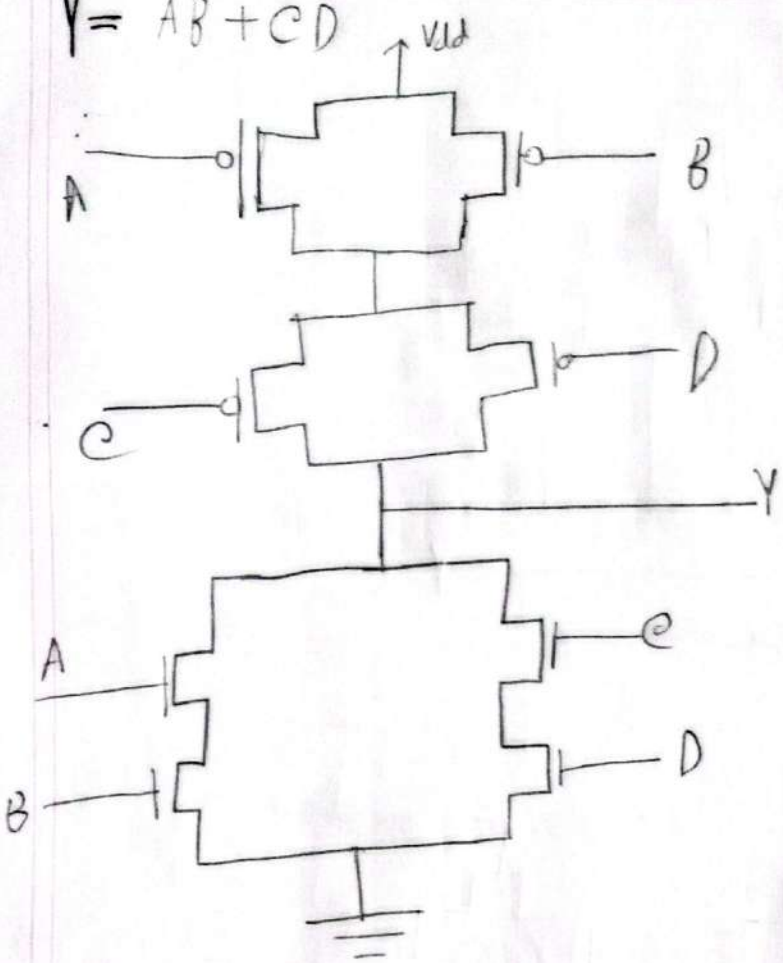
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



$Y = AB + C$

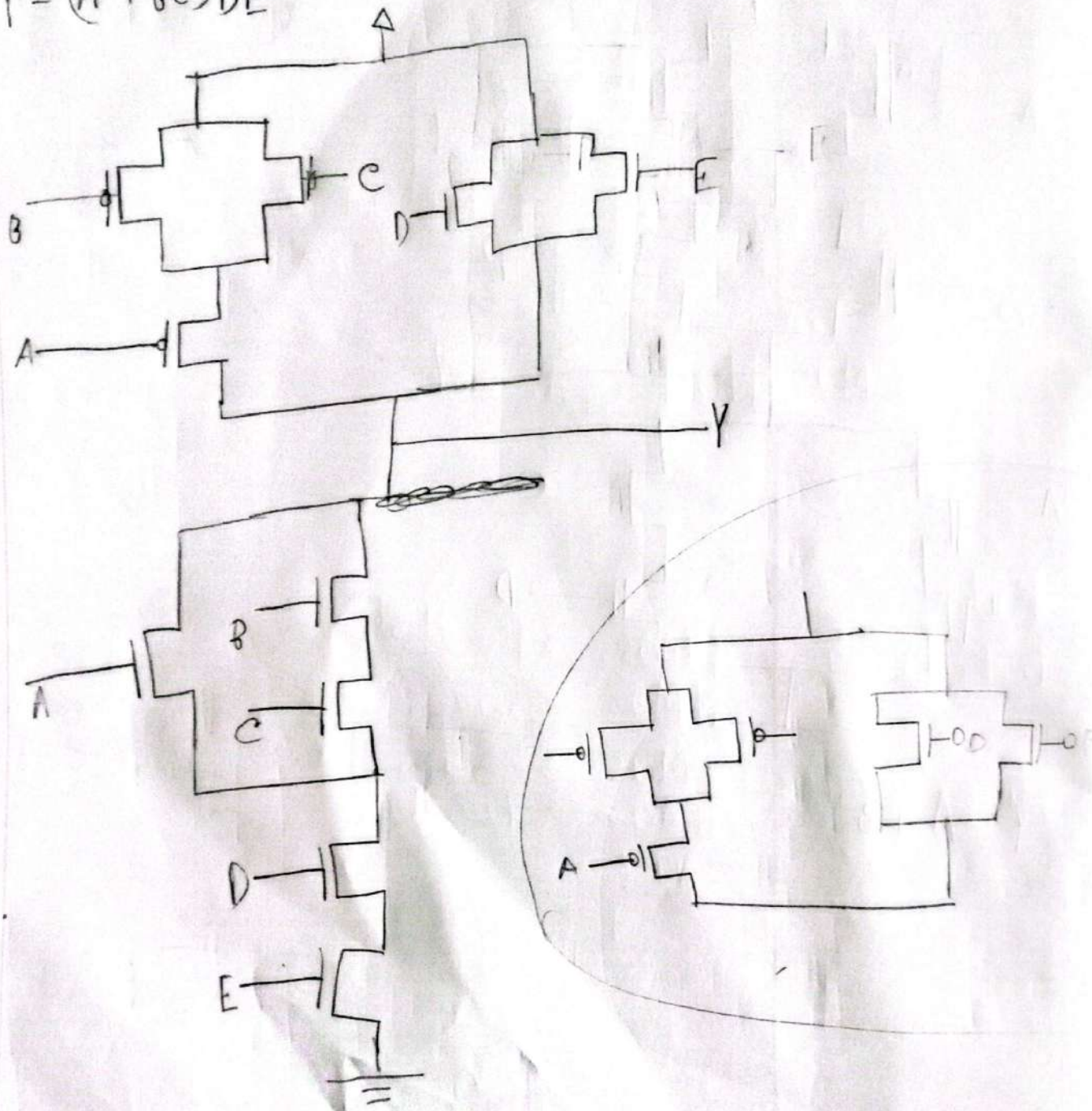


$Y = \overline{AB + CD}$



~~Y = A + B~~

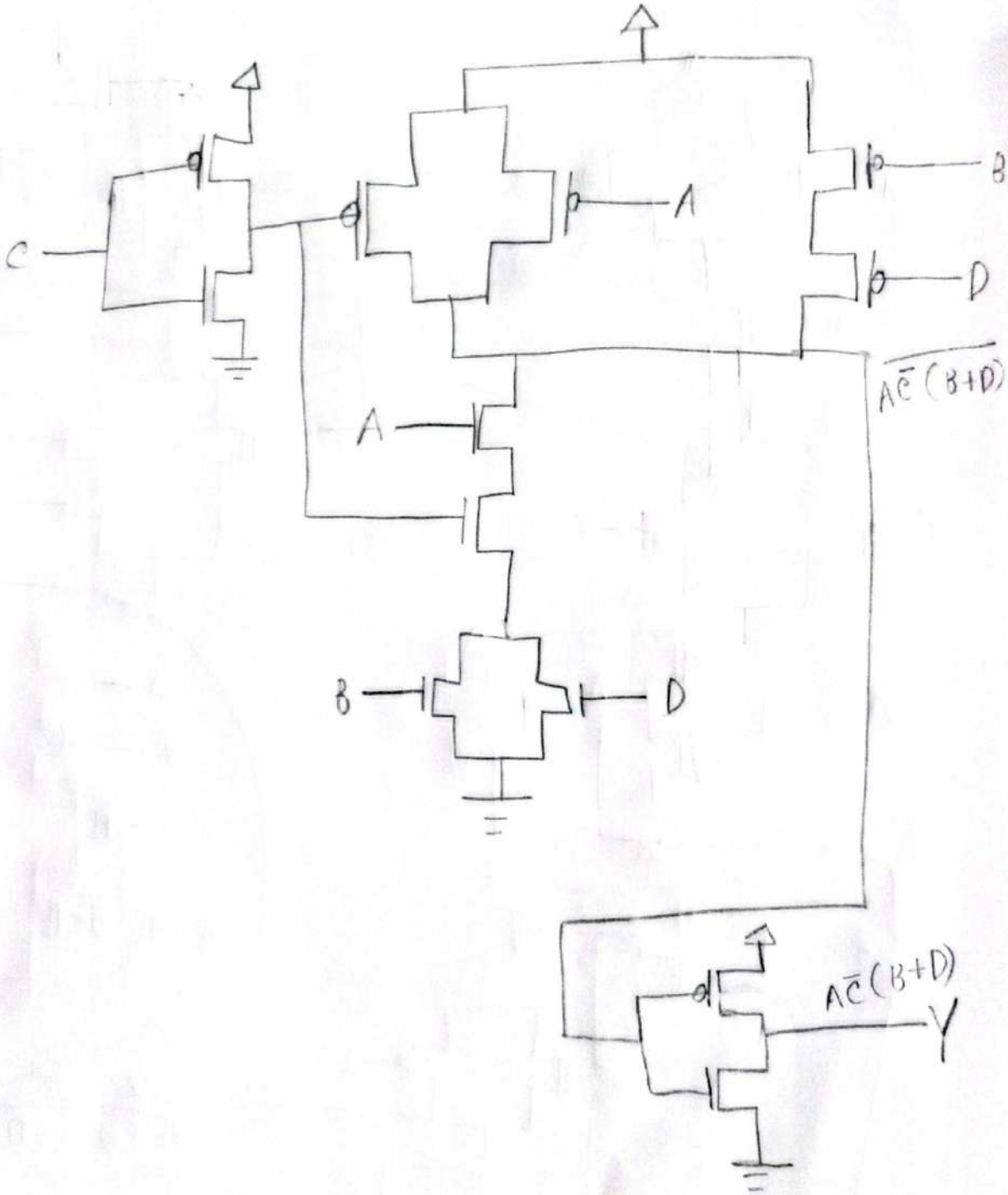
$Y = \overline{(A + BC)DE}$



next week এর monday তে CT-1 এগুলন পর্যন্ত যা যা পড়াইছে

$Y = (AB+D)A\bar{C} = AB \cdot A\bar{C} + DA\bar{C}$

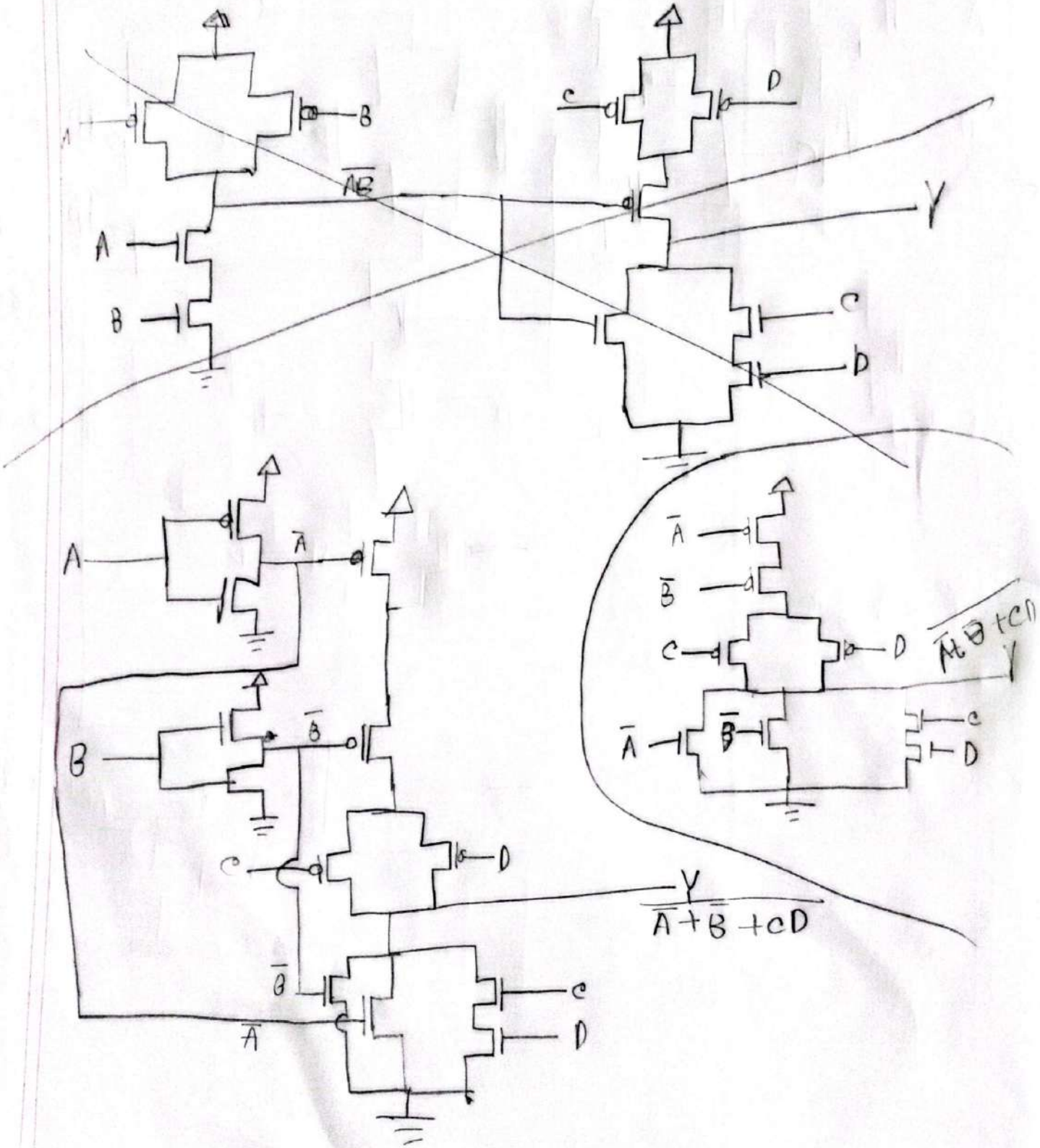
$= AB\bar{C} + AD\bar{C} = A\bar{C}(B+D)$



H.W $F = \overline{D + A(B+C)}$

$F = A\bar{B} + \bar{B}CD$

$Y = \overline{\overline{A\bar{B} + CD}} = \overline{\bar{A} + \bar{B} + CD}$



Mid ও electronic ও Machine (50/50)
electronics সম্পূর্ণ compelete (17 August 2024)

18/08/2026

Slide ৪ 3

Transformer

→ A transformer is a static piece of apparatus by which ac electric power of ~~one~~ one voltage level in one circuit is transferred to another circuit at another voltage level maintaining the same frequency

→ Transformer works according to principle of faraday's law of electromagnetic induction

→ B
- ✓ Transfers electric power from one circuit to another.
- ✓ It does so without a change in frequency
- ✓ It accomplishes this

লেখার আগে slide changed

Slide ৪ 4
1885 (পুরোন slide এ 1985 লিখেছে)
Transfer invented

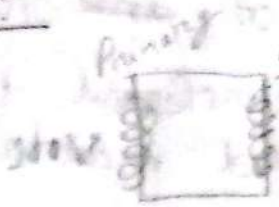
Slide ৪ 5
Variation এগুলোই Transformer এর

Slide ৪ 6
DC power was mainly is used in the 1880's and it was hard to transmit over distance
DC তে tranformer বাজে কাজে লাগে না। সুবিধে নেই।
তাই AC use হয়।

Slide 6: (কিছু নতে নিও। বাট ... তবে না করলে বড়ো ক্ষোভে আছে)

Slide 7: [illegible] সার্কিট

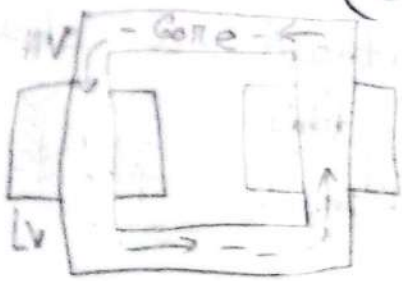
Slide 8:



Transformer এর input side fixed না।
(Primary, Secondary) = (input, output)

Tranforme দুই type

1. Core-type transformer (দুই দা L)
2. Shell-type transformer (E এর মাঝে I)



Core-type Construction



Shell-type construction

Faraday's law of electromagnatic induction

Magnatic flux ↑ magnatism	electron ↑ electricity

Slide 10

Lickege flux

Slide 11

এই অংশ বাদে পড়ার দরকার নাই।

Slide 16

E, I এর ছবি পাবেন Slide এ

24/8/2026

L 2 - Transformer.pdf

Slide 22

Ideal Transformer (Loss নাই) (100% efficient)

~~Static (কোন Mechanical sector নাই)~~

Slide 23

i.

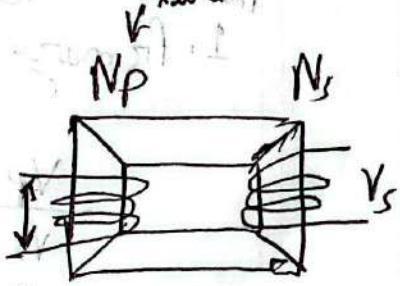
(ii)

(iii) Core-loss zero

(iv) core has infinite permeability.

(Magnetic flux flow হবে কিন্তু ক্ষতি হবে না)

Slide 24 P - Primary, S - secondary



$$\frac{V_p(t)}{V_s(t)} = \frac{N_p}{N_s} = a = \frac{I_s}{I_p}$$

Voltage

$$N_p i_p(t) = N_s i_s(t)$$

Slide 25

$$\frac{V_p}{V_s} = a \qquad \frac{I_p}{I_s} = \frac{1}{a}$$

~~Slide 28~~

Slide 28

angle

$\theta_p = V_p - I_p$ $\quad P = VI\cos(\theta_V - \theta_I)$

$$P_{in} = V_p I_p \cos\theta_p$$

$$P_{out} = V_s I_s \cos\theta_s$$

$$P_{out} = \eta \cdot P_{In}$$

Textbook of electrical Technology

~~Text~~ Book Slide ~~No~~ 267 - 32.1

The maximum flux density in the core of a 250/3000 volts, ~~50 Hz single-pha~~ E.M.F per turn is 8 volt,

Primary ↑ Secondry

1. Primary and Secondary turns -

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{250}{3000} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{1}{12} = \frac{N_p}{N_s}$$

$$\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{1}{12}$$

$$\Rightarrow \frac{I_p}{I_s} = 12$$

Slide 32

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

$K > 1 \rightarrow$ Step-up transformer

$K < 1 \rightarrow$ step-~~down~~ down transformer

$K = 1 \rightarrow$ Ideal transformer

Book. 267 – 32.1 (i)

$E_1 = N_1 \times$ e.m.f induced/turn

$$\Rightarrow N_1 = \frac{E_1}{\text{e.m.f induce/turn}}$$

turns সর্বদা ভিত্তি না

$$\Rightarrow N_1 = \frac{250}{8} = 31.25 = 32$$

$$N_2 = \frac{3000}{8} = 375$$

32.2 Voltage 11000/500V ; Emf 10.4 ← ধ্রুব নিন

$$N_1 = \frac{11000}{10.4} = 1058$$

$$N_2 = \frac{500}{10.4} = 49$$

$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{500}{11000} = \frac{1}{20}$$

$$\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{49}{1058} = 0.046$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{1058}{49} = 2$$

15/5/2026

Book Page 272

1. The number of turns on the primary and secondray of a 1−ϕ tranformer are 350 and 35 respectively. If primary is connected to a 2.2kV, 50Hz suply, determine the secondry voltage on load.

We know,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2}$$

$$\Rightarrow V_2 = \frac{N_2}{N_1} \times V_1$$

$$= \frac{35}{350} \times 2200$$

$$= 220\,V$$

Primary $I = 5A$

Secondry $= I_2$

$$\frac{I_2}{I_{o1}} = \frac{V_1}{V_2} = \frac{220}{22} = 10$$

$$\Rightarrow I_2 = \frac{1}{10} \times I_1$$

$$I_2 = \frac{1}{10} \times 5$$

$$= 50\,A$$

4(a) A 40 kVA, 3,300/240V, 50 Hz, 1 phase transformer has 660 turns on the primary.

⇒ Determine the number of turns on secondary.

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{I_s}{I_p}$$

Motor Inductive Load

Far

We know, $\frac{N_1}{N_2} = \frac{V_1}{V_2}$

$$\Rightarrow \frac{660}{N_2} = \frac{3300}{240}$$

$$\Rightarrow N_2 = \frac{240}{3300} \times 660$$

$$= 48$$

2(b) A 3000/200 V, 50 Hz, 1 phase transformer ..... area of 150 cm^2 and has 80 turns in the low-voltage welding. Calculate num turn in high-voltag

We know,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2}$$

$$\Rightarrow \frac{N_1}{80} = \frac{3000}{200}$$

$$\Rightarrow N_1 = \frac{3000}{200} \times 80 = 1200$$

Active power
Reactive power
Apparent power

Volt Ampere Reactive

Volt Ampere

$P = VI\cos\theta$ W

1 KWh

100 W X 10 h

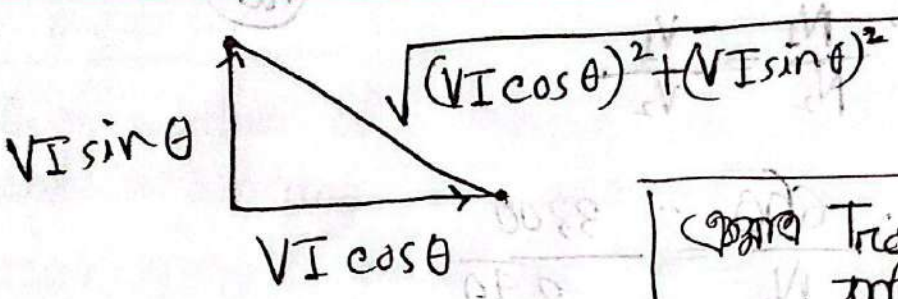
$P = VI \sin\theta$ VAR

$P = VI$ VA

Power ত্রিভুজ

ডিগ্রি - θ Real (Active)
বাতাস - Reactive

Reactive মাপার ~~Act~~ Real ও flow করাও আরামের করার



কারণ Transformer VA তে
মাপি এবং Watt এ হয়

Why tranformers are rated in VA not in Watt

7 september CT-2

31/8/2026

X-reactance

Slide 40

(I) Core or iron loss

Slide 41

Copper Loss (Heating loss I^2R)

Slide 58

Equivalent circuit of transformer

Resistor ও resistance বলতে

Inductor " " বলতে Inductance ও বোঝায়।

Ω

H

Henry to ohm

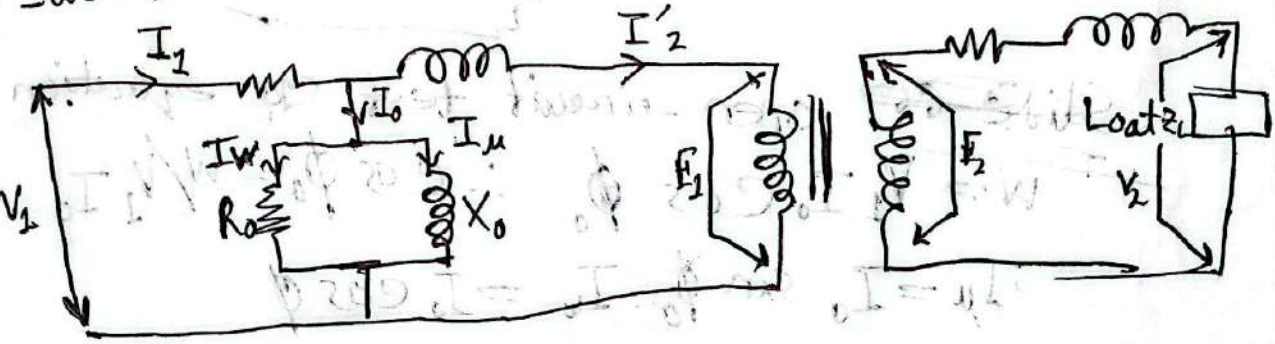
$2\pi fL = \omega L$

$= X_L$

Ω

Primary – secondary

I_w - working



I_0 - No load current

I_w - working component

I_μ - Magnetizing component

variac = variable ac

wattmeter

Slide 59 → simplified
Slide 60 → simplied
} এই তিনটি circuit নিয়ে আলোচনা

Slide 61

Slide 63

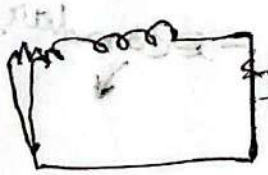
Transformer এর rating VA তে প্রকাশ করা হয়। কেন প্রকাশ করা হয়?

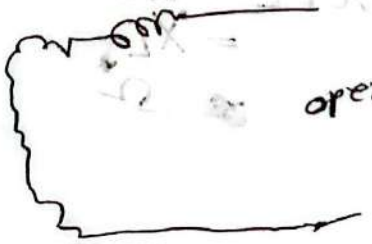
Slide 64

open-circuit test → High তে supply দেয়া হয়
short " → Low voltage side short করা হয়

Slide 65/66

open-circuit test → High voltage side open থাকে, Low তে supply





এখানে কোন প্রবাহ হবে না

Slide 66

open-circuit test এর equation

$W = V_1 I_0 \cos\phi_0 \quad \therefore \cos\phi_0 = W/V_1 I_0$

$I_\mu = I_0 \sin\phi_0 \quad I_w = I_0 \cos\phi$

$\therefore X_0 = V_1/I_\mu$ and $R_0 = V_1/I_w$

আজকে পর্যন্ত CT ভিত্তিক্যাক Mutual ইন্ডাক্টেন্স ও Transformer related Basic que

Open circuit test

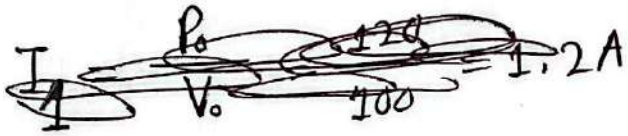
$V_o = 100\ V$

$I_o = 2\ A$

$P_o = 120W$

$R_o = ?$

$X_o = ?$





$$\cos\phi_o = \frac{P_o}{V_o I_o} = \frac{120}{100 \times 2} = 0.6 \quad \therefore \phi_o = 53.13$$

$$I_u = I_o \sin\phi_o = 2 \times \sin(53.13) = 1.6$$

$$I_w = I_o \cos\phi = 2 \times 0.6 = 1.2$$

$$X_o = \frac{V_o}{I_u} = \frac{100}{1.6} = 62.5$$

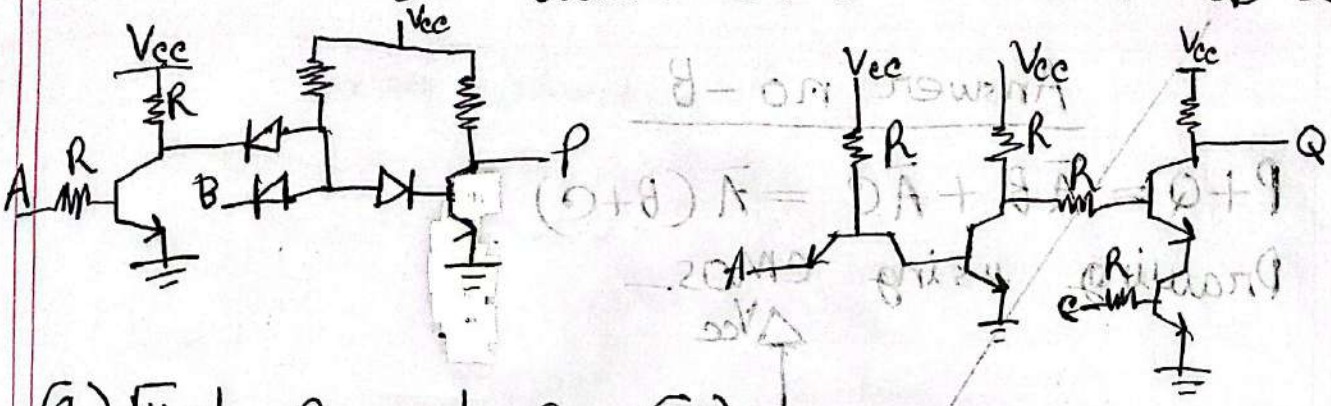
$$R_o = \frac{V_o}{I_w} = \frac{100}{1.2} = 83.33$$

CamScanner

56 Batch sec C CT-1

1. In active mode, the base-emitter junction is forward biased.
2. The current gain of BJT is represented by the symbol β.
3. For amplication, the BJT operates in the active region.
4. BJT has three terminals : Emitter, Base and Collector.
5. The relation between current in BJT is $I_E = I_C + I_B$.
6. The input junction of BJT, is Base-Emitter
7. RTL has relatively low noise margin compared to modern logic families.
8. RTL circuits use resistors and transistors as components.
9. Compared to RTL, DTL has higher noise margin
10. DTL does not use transistor. (True/False)
    Ans: False

56 Batch CT-1 section (D) set (B)



(a) Find P and Q (5)

(b) Draw a circuit for P+Q using CMOS (15)

Answer to the question no-a

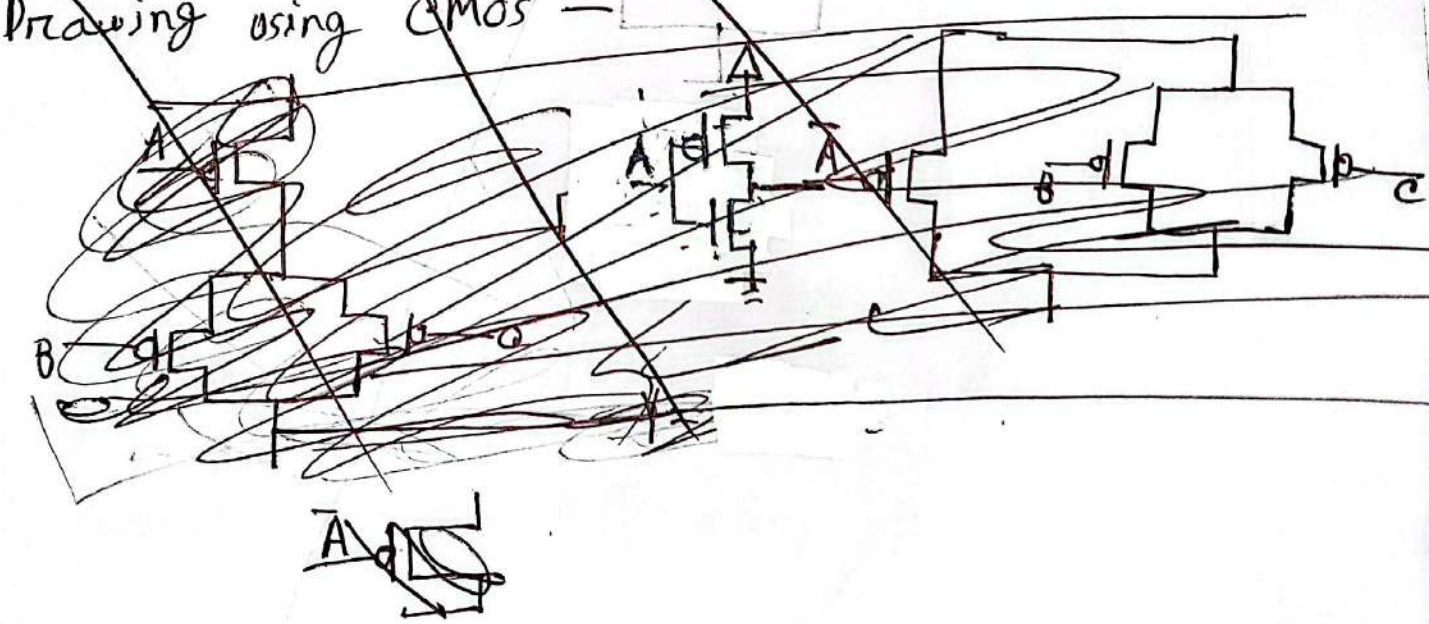
$P = \bar{A}B$

$Q = \bar{A}C$

Answer no-b

$P+Q = \bar{A}B + \bar{A}C = \bar{A}(B+C)$

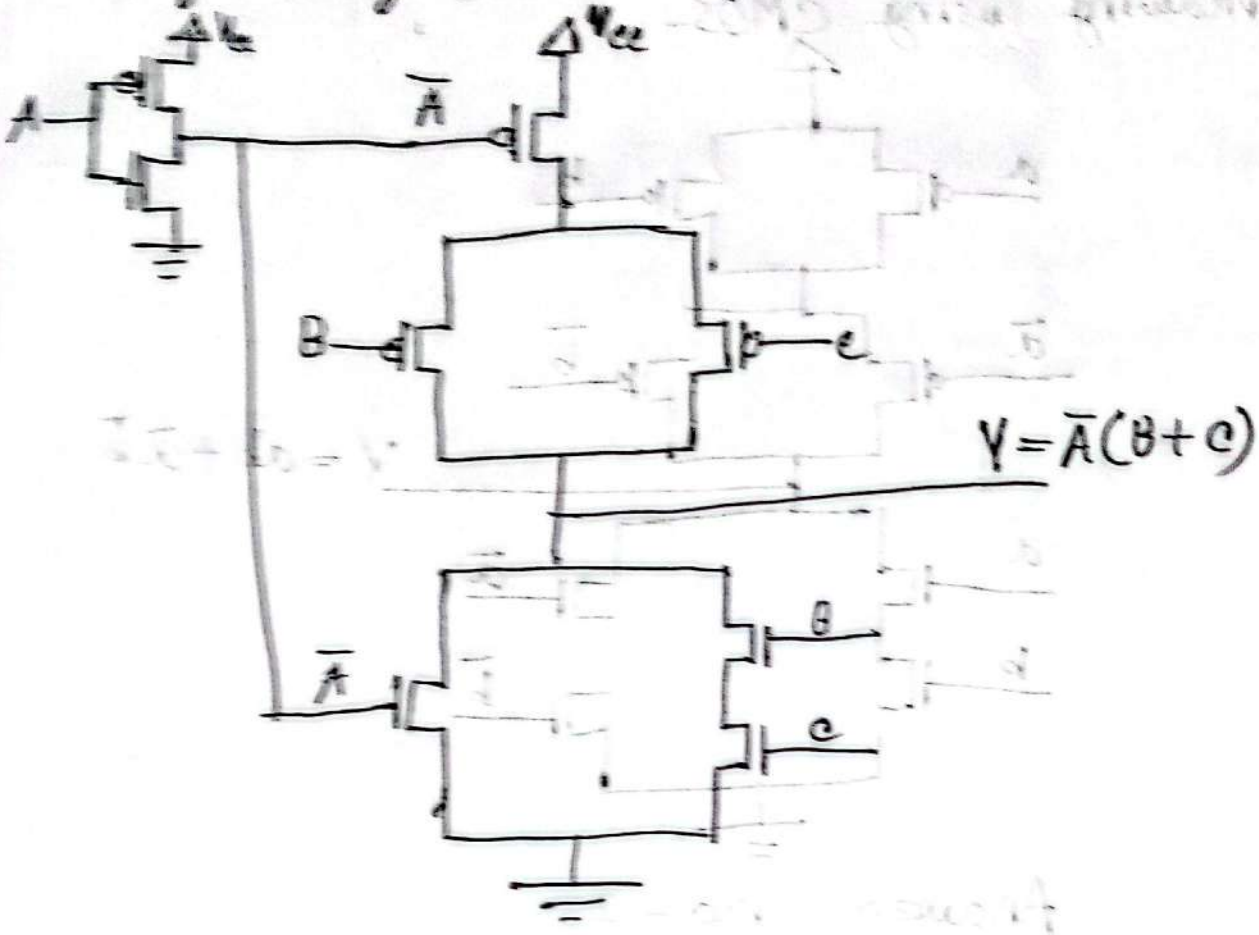
Drawing using CMOS —



Answer no-B

$P+Q = \bar{A}B + \bar{A}C = \bar{A}(B+C)$

Drawing using CMOS-



56th CT-1 section D ~~set A~~

Question 1:

Implement an XNOR gate using CMOS

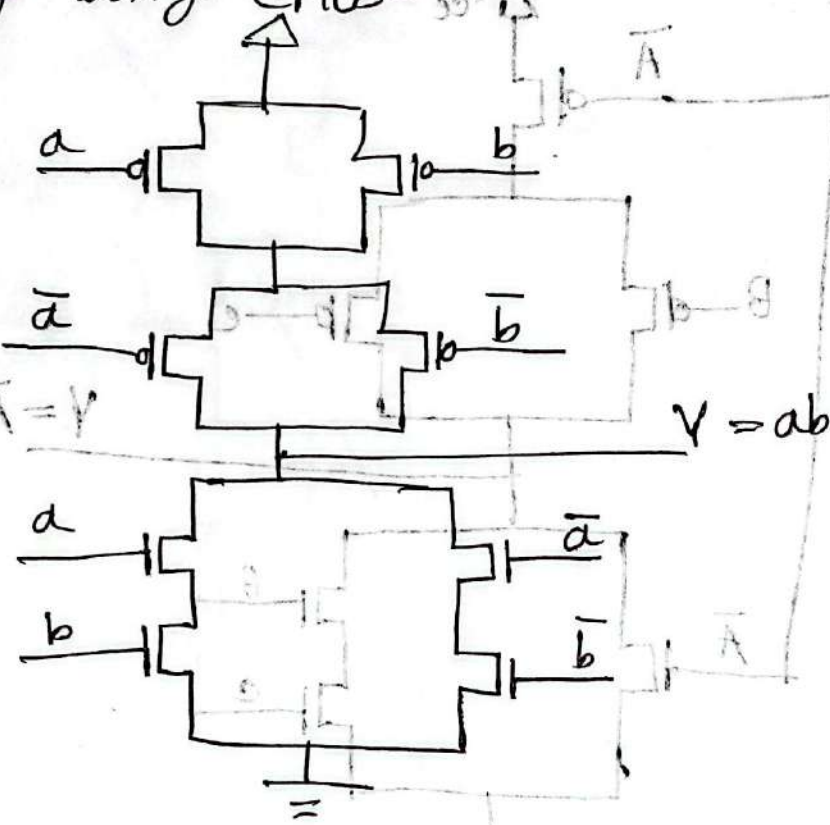
Question 2:

Implement $Y = A + B\bar{C}$ using DTL

Answer to the question no-1

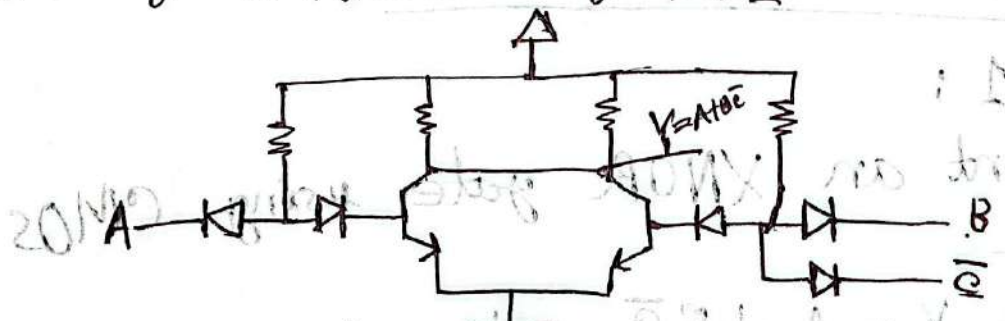
For XNOR gate $Y = ab + \bar{a}\bar{b}$

Drawing using CMOS-



Answer no - 2:

Drawing $A + B\bar{C}$ using DTL



55 Batch section D

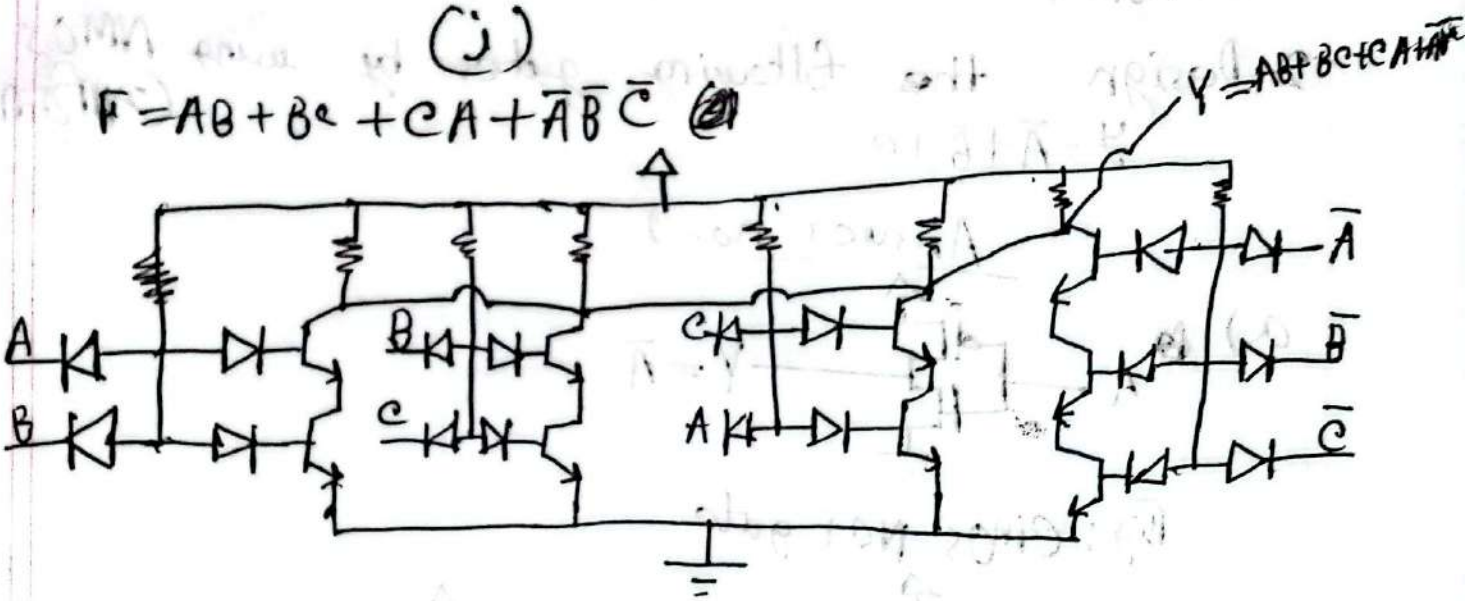
1. Design the Basic Gates using

(i) $F = AB + BC + CA + \bar{A}\bar{B}\bar{C}$ (DTL)

(ii) $F = \bar{A} + \bar{B} + \bar{C} + \overline{AB}$ (RTL)

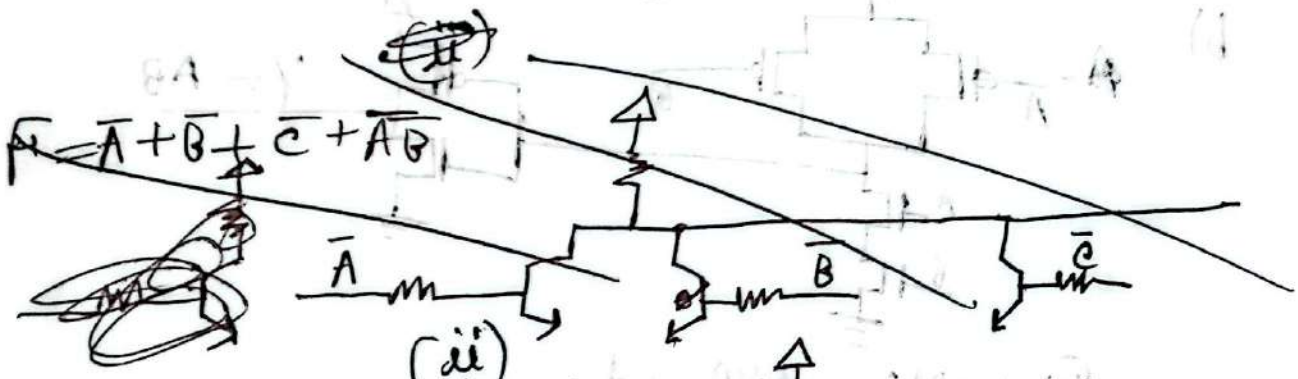
(i)

$F = AB + BC + CA + \bar{A}\bar{B}\bar{C}$



(ii)

$F = \bar{A} + \bar{B} + \bar{C} + \overline{AB}$

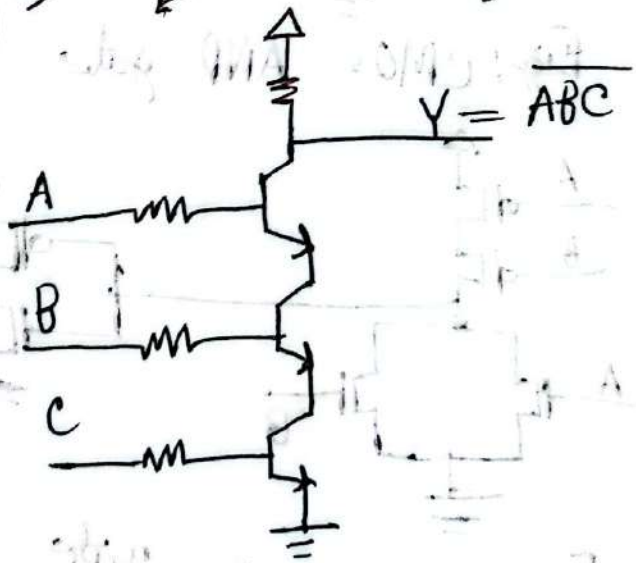


$F = \bar{A} + \bar{B} + \bar{C} + \overline{AB}$

$= \bar{A} + \bar{B} + \bar{C} + \bar{A} + \bar{B}$

$= \bar{A} + \bar{B} + \bar{C}$

$= \overline{ABC}$



59 Batch section-C

1. Design the following gates by using CMOS
   a) NOT gate
   b) AND gate
   c) OR gate
   d) XOR gate

2. Design the following gates by using NMOS (only)
   $Y = \bar{A} + \bar{B} + \bar{C}$

Answer no-1

a)

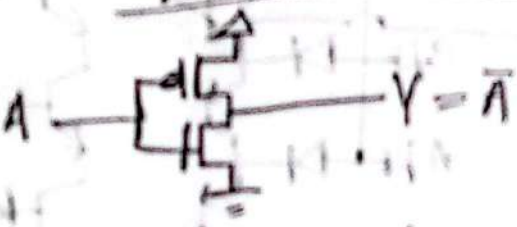


Fig: CMOS NOT gate

b)

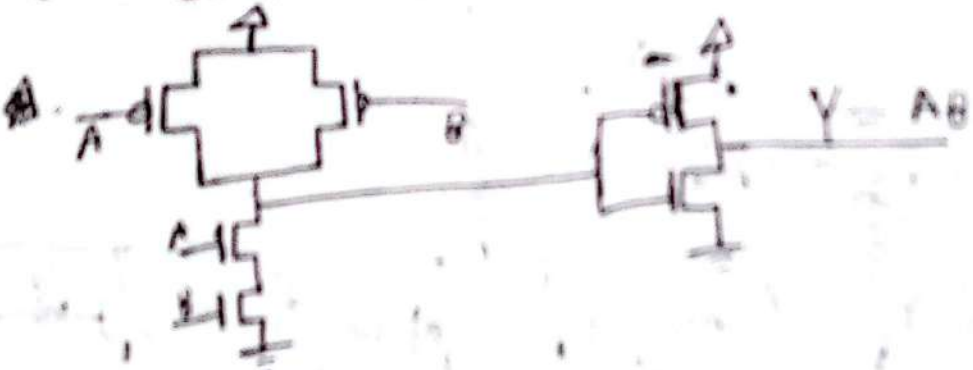


Fig: CMOS AND gate

c)

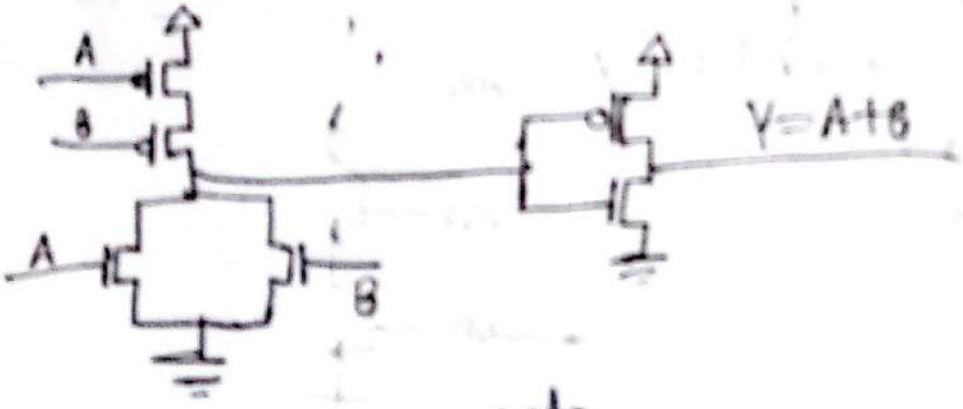
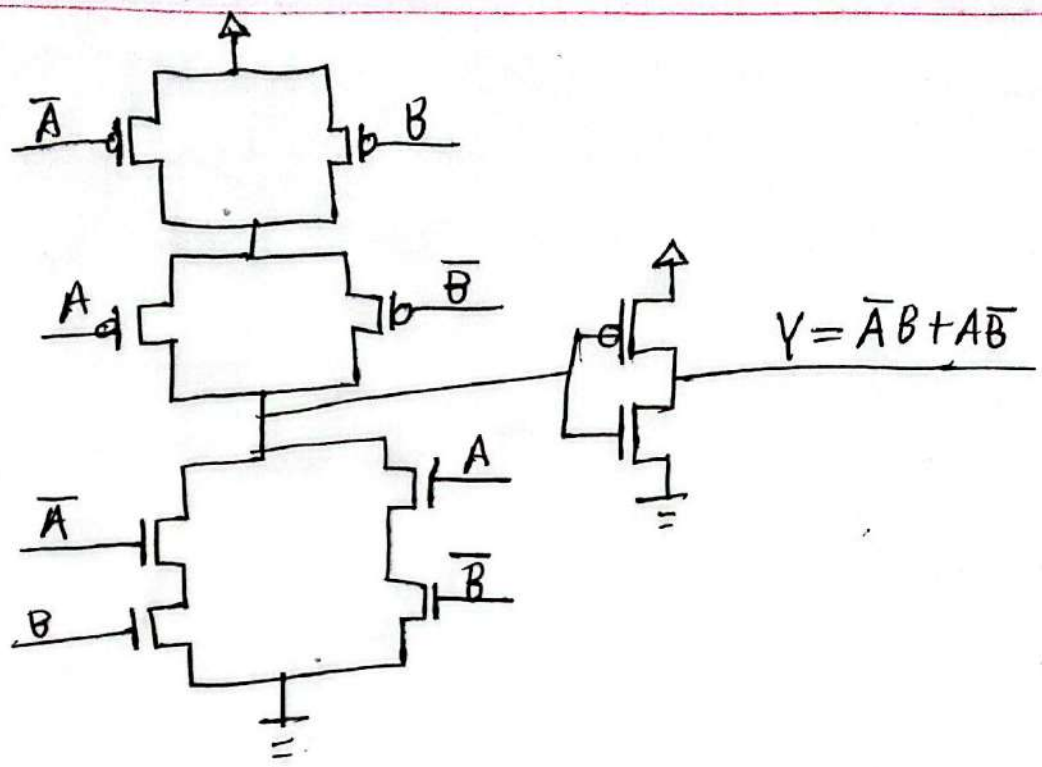


Fig: CMOS OR gate

d)



53 Batch section c

1. Desine the basic circuits by RTL, TTL and CMOS
2. Desigh the circuit $Y = ABC$

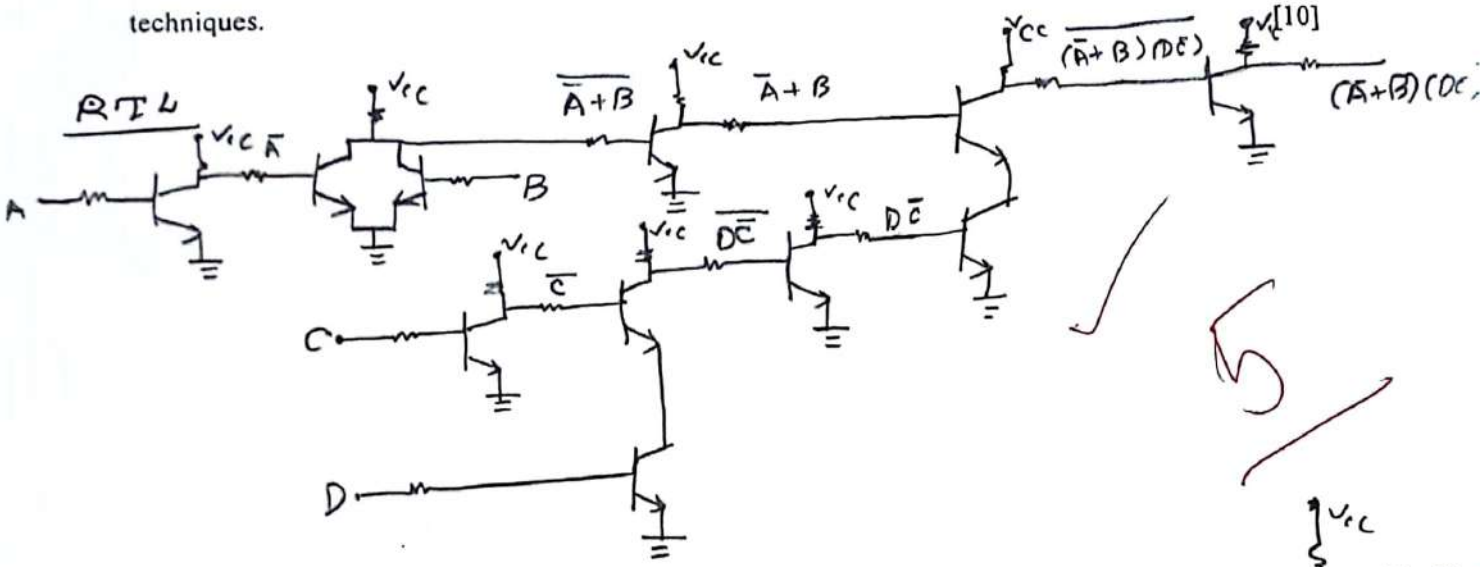
University of Asia Pacific
Department of Computer Science & Engineering
Class Quiz 1, Fall-2023

15

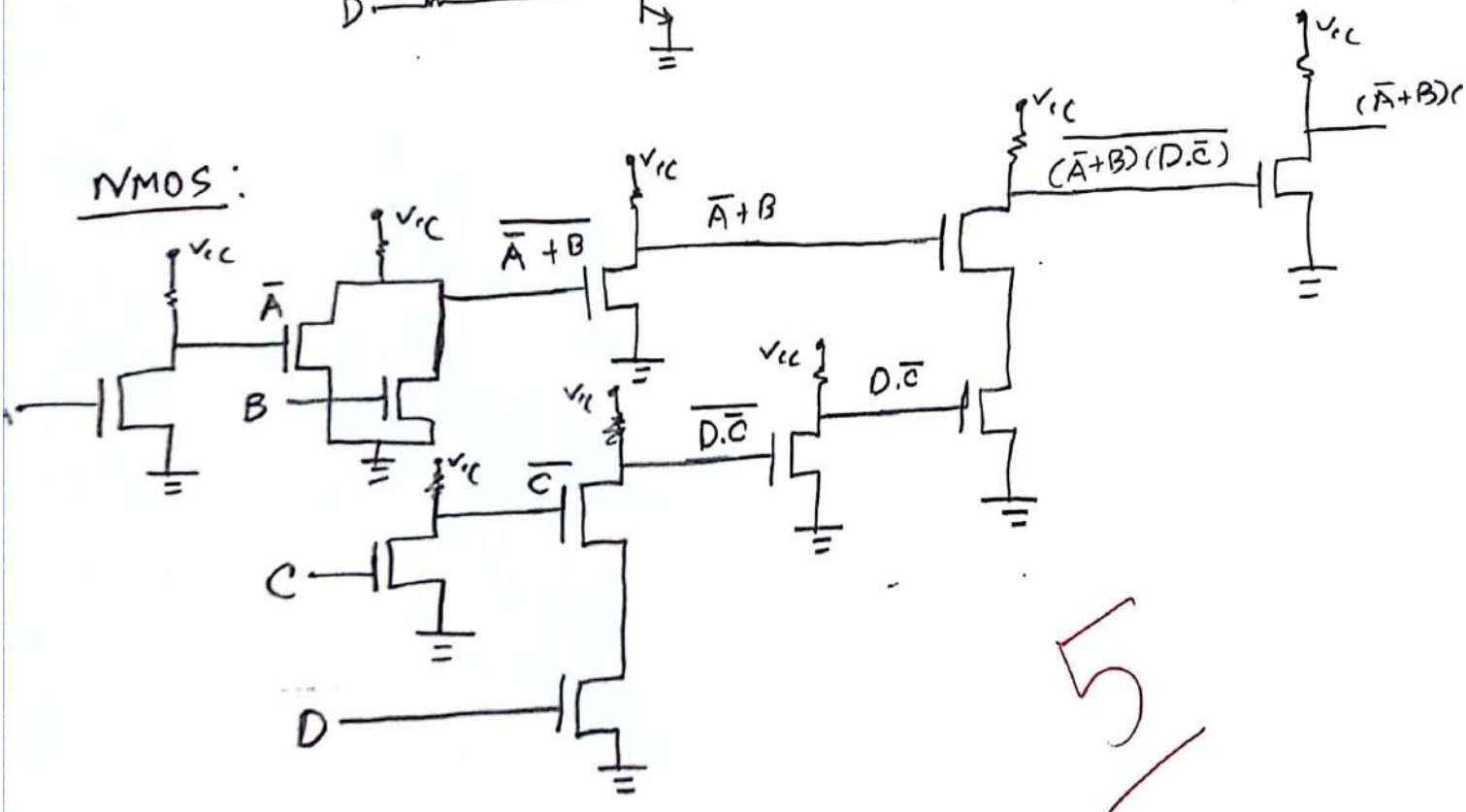
Course Title: Electronic Devices & Circuits | **Course Code: EEE 201**
Time: 30 min | **Full Marks: 15**

Name: MD. Sydul Islam Zubair | ID: 22201174

1. Construct the circuit for the logic function, $F = (\bar{A} + B)(D\bar{C})$ using RTL and NMOS design techniques. [10]



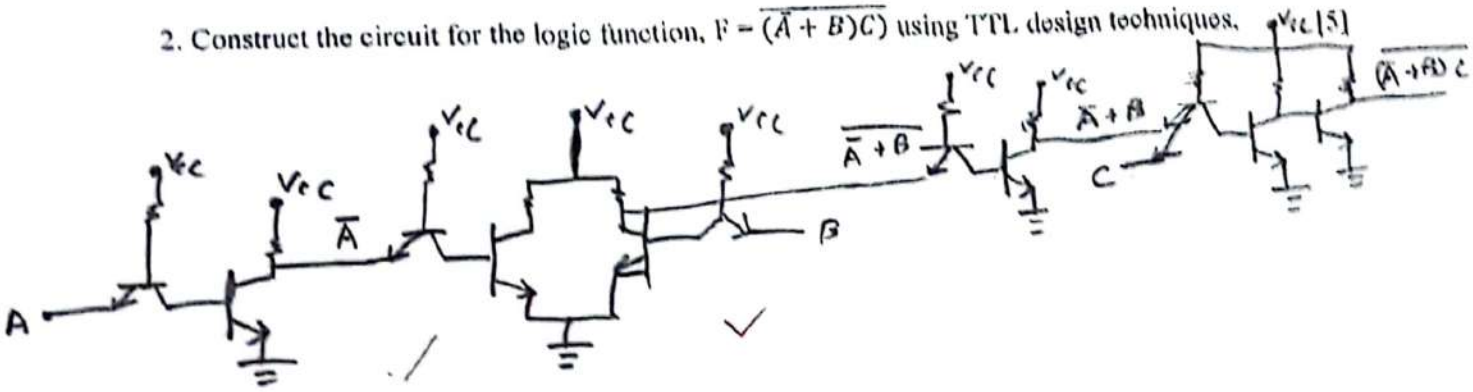
5



5

CamScanner

2. Construct the circuit for the logic function, $F = \overline{(\bar{A} + B)C}$ using TTL design techniques. [5]



5

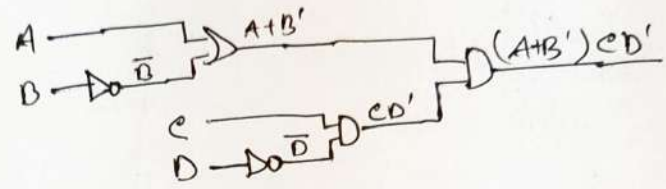
Md. Junaid Hossain
22201017

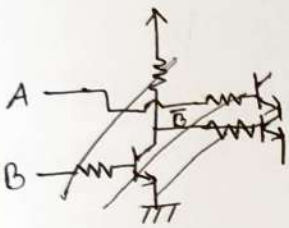
Quiz 1
EEE 201
Set B

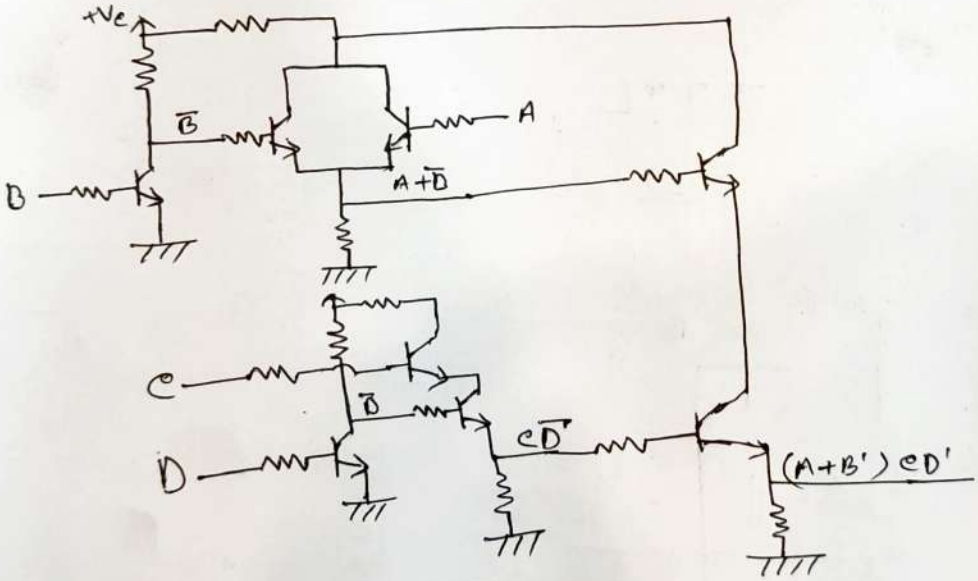
1. Design logic circuit using TTL and RTL for the expression Y= (A+B')D'C
2. Design logic circuit using CMOS for Y= A'B+CD

(1)

$(A+B')D'C$

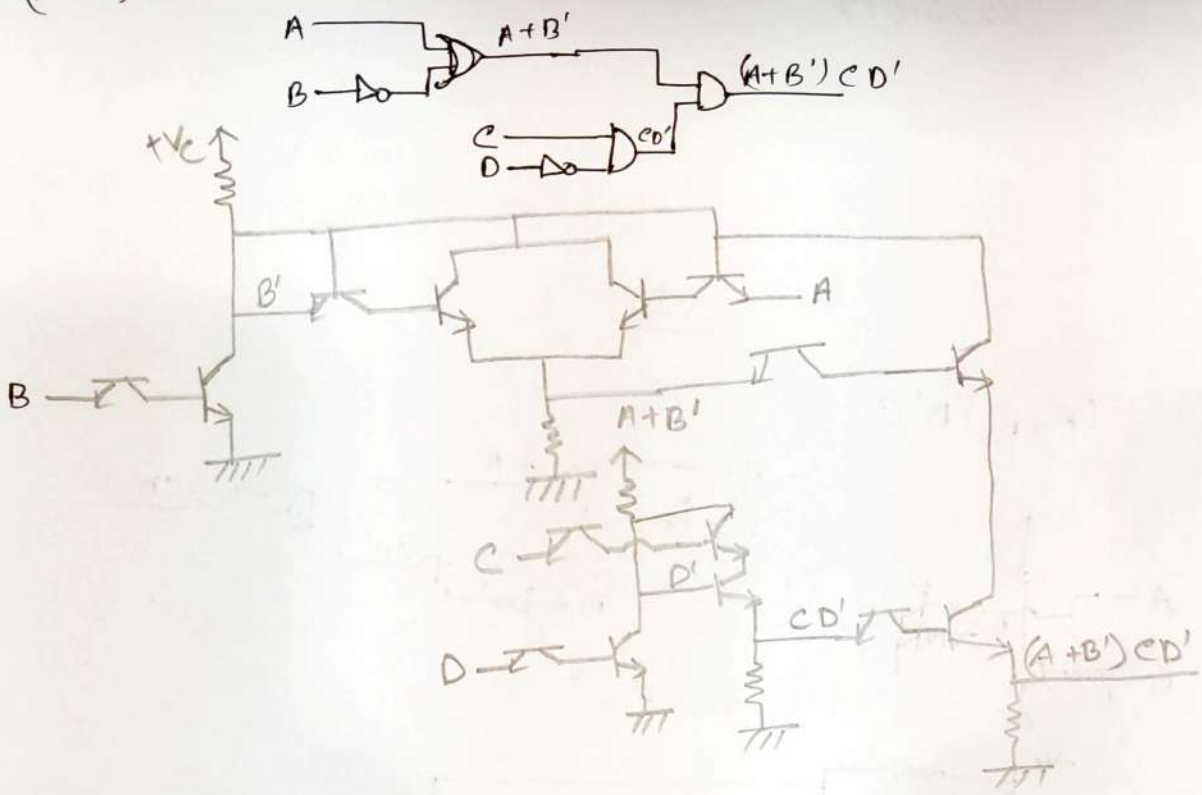






RTL

$(A+B')\,CD'$



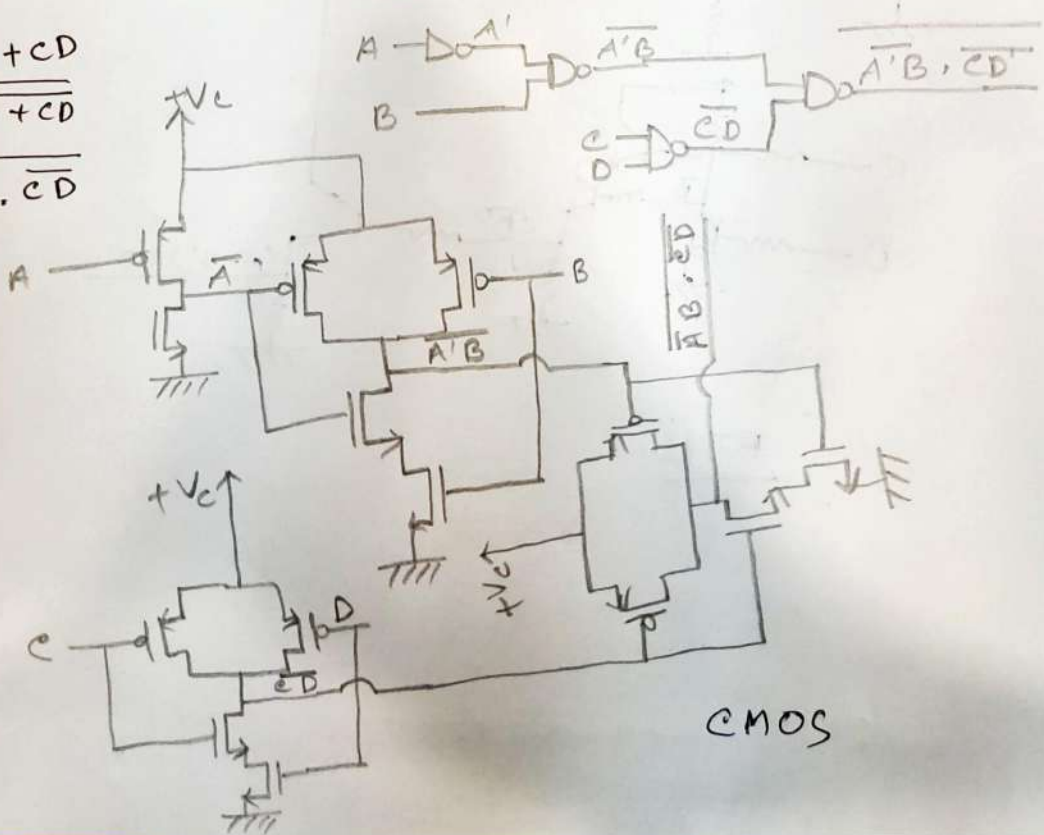
TTL

(2)

$A'B+CD$

$= \overline{\overline{A'B+CD}}$

$= \overline{\overline{A'B}\cdot\overline{CD}}$



CMOS

Quiz 1
EEE 201
Set A

Marzan Ahmed.
22201055

1. Design logic circuit using NMOS and DTL for the expression Y= (A'+B')DC'
2. Design logic circuit using CMOS for Y= (AB+C'D)'

① $Y = (A'+B')DC'$

$= \overline{(A'+B')DC'}$

$= \overline{(A'+B')} + \overline{DC'}$

NMOS:

